Trigonometric Substitutions and Inverse Trigonometric Expressions

Recovering algebraic expressions in x

This version contains worked solutions. Square roots denote the nonnegative principal square root, and every sign choice follows from the stated range of θ.

Reference sheet

$$\sin^2\theta+\cos^2\theta=1,\qquad 1+\tan^2\theta=\sec^2\theta,\qquad 1+\cot^2\theta=\csc^2\theta.$$

$$\sin(2\theta)=2\sin\theta\cos\theta,\quad \cos(2\theta)=1-2\sin^2\theta=2\cos^2\theta-1,\quad \tan(2\theta)=\frac{2\tan\theta}{1-\tan^2\theta}.$$

Principal ranges used below are

$$\arcsin u\in[-\pi/2,\pi/2],\qquad \arccos u\in[0,\pi],\qquad \arctan u\in(-\pi/2,\pi/2).$$

For arcsec problems with positive argument restrictions, use $\theta\in[0,\pi/2)$. For unrestricted arcsec, use $\theta\in[0,\pi]$ with $\sin\theta\ge 0$. For arccsc, use $\theta\in[-\pi/2,\pi/2]$ with $\cos\theta\ge 0$, and for arccot, use $\theta\in(0,\pi)$.

1 Direct Substitution Recovery

Use the stated substitution and angle range. These are the conversions that often occur at the final stage of a trigonometric-substitution integral.

1. $x=3\sin\theta$, where $-\pi/2\le\theta\le\pi/2$. Express $2\cos\theta$ in terms of x.

Solution. From $\sin\theta=x/3$,

$$\cos\theta=\sqrt{1-\sin^2\theta}=\sqrt{1-\frac{x^2}{9}}=\frac{\sqrt{9-x^2}}{3}.$$

Cosine is nonnegative on the stated range, so

$$\boxed{2\cos\theta=\frac{2\sqrt{9-x^2}}{3}},\qquad |x|\le 3.$$

2. $x=5\tan\theta$, where $-\pi/2<\theta<\pi/2$. Express $\frac{1}{2}\sin\theta$ in terms of x.

Solution. Here $\tan\theta=x/5$ and $\sec\theta>0$. Thus

$$\sec\theta=\sqrt{1+\tan^2\theta}=\frac{\sqrt{x^2+25}}{5}.$$

Since $\sin\theta=\tan\theta/\sec\theta$,

$$\boxed{\frac{1}{2}\sin\theta=\frac{x}{2\sqrt{x^2+25}}}.$$

3. $x = 2\sec\theta$, where $0 \le \theta < \pi/2$. Express $\frac{1}{4}\sin\theta$ in terms of x.
Solution. We have $\cos\theta = 2/x$. Since sine is nonnegative,

$$\sin\theta = \sqrt{1 - \frac{4}{x^2}} = \frac{\sqrt{x^2-4}}{x}.$$

Therefore

$$\boxed{\frac{1}{4}\sin\theta = \frac{\sqrt{x^2-4}}{4x}}, \qquad x \ge 2.$$

4. $x = 4\sin\theta$, where $-\pi/2 < \theta < \pi/2$. Express $\tan\theta$ in terms of x.
Solution. We have $\sin\theta = x/4$ and $\cos\theta = \sqrt{16-x^2}/4 > 0$. Therefore

$$\boxed{\tan\theta = \frac{x}{\sqrt{16-x^2}}}, \qquad |x| < 4.$$

5. $x = 3\tan\theta$, where $-\pi/2 < \theta < \pi/2$. Express $\sec\theta$ in terms of x.
Solution. Because $\sec\theta > 0$,

$$\sec\theta = \sqrt{1+\tan^2\theta} = \sqrt{1 + \frac{x^2}{9}}.$$

Hence

$$\boxed{\sec\theta = \frac{\sqrt{x^2+9}}{3}}.$$

6. $x = 6\sec\theta$, where $0 \le \theta < \pi/2$. Express $\tan\theta$ in terms of x.
Solution. Since $\sec\theta = x/6$ and $\tan\theta \ge 0$,

$$\boxed{\tan\theta = \sqrt{\sec^2\theta - 1} = \frac{\sqrt{x^2-36}}{6}}, \qquad x \ge 6.$$

7. $x = 2\sin\theta$, where $0 < \theta \le \pi/2$. Express $3\csc\theta$ in terms of x.
Solution. Since $\sin\theta = x/2$,

$$\boxed{3\csc\theta = \frac{3}{\sin\theta} = \frac{6}{x}}, \qquad 0 < x \le 2.$$

8. $x = 7\tan\theta$, where $-\pi/2 < \theta < \pi/2$. Express $4\cos\theta$ in terms of x.
Solution. We have $\sec\theta = \sqrt{x^2+49}/7$. Therefore

$$\boxed{4\cos\theta = \frac{4}{\sec\theta} = \frac{28}{\sqrt{x^2+49}}}.$$

9. $x = 5\sec\theta$, where $0 < \theta < \pi/2$. Express $2\cot\theta$ in terms of x.
Solution. The identity $\tan^2\theta = \sec^2\theta - 1$ gives $\tan\theta = \sqrt{x^2-25}/5$. Hence

$$\boxed{2\cot\theta = \frac{10}{\sqrt{x^2-25}}}, \qquad x > 5.$$

2 Reciprocal Compositions

Start by setting θ equal to the inverse trigonometric expression. These problems require little more than taking a reciprocal, but domains still matter.

10. $5\csc\left(\arcsin\dfrac{x}{5}\right)$

Solution. $\sin\theta = x/5$, so $\csc\theta = 5/x$ and the expression is $\dfrac{25}{x}$ for $0 < |x| \le 5$.

11. $\sec(\arccos 3x)$

Solution. $\cos\theta = 3x$, so the value is $\dfrac{1}{3x}$ for $0 < |x| \le 1/3$.

12. $2\cot(\arctan 2x)$

Solution. $\tan\theta = 2x$, so $\cot\theta = 1/(2x)$ and the value is $\dfrac{1}{x}$ for $x \neq 0$.

13. $\cos\left(\operatorname{arcsec}\dfrac{x}{4}\right)$

Solution. $\sec\theta = x/4$, so the value is $\dfrac{4}{x}$ for $|x| \ge 4$.

14. $\tan(\operatorname{arccot} 5x)$

Solution. $\cot\theta = 5x$, giving $\dfrac{1}{5x}$ for $x \neq 0$. At $x = 0$ we have $\theta = \pi/2$ and the tangent is undefined, so the point must be excluded even though arccot is defined there.

3 One Pythagorean Step

Use one Pythagorean identity, then choose the sign from the range of θ.

15. $3\cos\left(\arcsin\dfrac{x}{3}\right)$

Solution. $\theta \in [-\pi/2, \pi/2]$ so $\cos\theta \ge 0$: $\cos\theta = \sqrt{1 - x^2/9}$ and the value is $\sqrt{9 - x^2}$ for $|x| \le 3$.

16. $\sin(\arccos 4x)$

Solution. $\theta \in [0, \pi]$ so $\sin\theta \ge 0$: the value is $\sqrt{1 - 16x^2}$ for $|x| \le 1/4$.

17. $\tan\left(\arcsin\dfrac{x}{2}\right)$

Solution. $\sin\theta = x/2$, $\cos\theta = \sqrt{4 - x^2}/2 > 0$, so $\dfrac{x}{\sqrt{4 - x^2}}$ for $|x| < 2$.

18. $5\sec\left(\arctan\dfrac{x}{5}\right)$

Solution. $\sec^2\theta = 1 + x^2/25$ and $\sec\theta > 0$ on $(-\pi/2, \pi/2)$, so the value is $\sqrt{25 + x^2}$ for all x.

19. $\cos(\arctan 4x)$

Solution. $\dfrac{1}{\sqrt{1 + 16x^2}}$ for all x.

20. $\cot\left(\arccos\dfrac{x}{6}\right)$

Solution. $\cos\theta = x/6$, $\sin\theta = \sqrt{36 - x^2}/6 \ge 0$, so $\dfrac{x}{\sqrt{36 - x^2}}$ for $|x| < 6$. The sign of the answer follows the sign of x automatically; no absolute value appears.

21. $\csc(\arccos 2x)$

Solution. $\dfrac{1}{\sqrt{1 - 4x^2}}$ for $|x| < 1/2$.

22. Simplify $3\cos(\arcsin(x/4))$.
Solution. Let $\theta = \arcsin(x/4)$. Then $\sin\theta = x/4$ and $\cos\theta \ge 0$, so

$$\boxed{3\cos(\arcsin(x/4)) = \frac{3\sqrt{16-x^2}}{4}}, \qquad |x| \le 4.$$

23. Simplify $2\sin(\arctan(3x))$.
Solution. Let $\theta = \arctan(3x)$. Since $\tan\theta = 3x$ and $\sec\theta = \sqrt{1+9x^2}$,

$$\boxed{2\sin(\arctan(3x)) = \frac{6x}{\sqrt{1+9x^2}}}.$$

24. For $0 < x \le \frac{1}{2}$, simplify $\frac{1}{5}\tan(\arccos(2x))$.
Solution. Let $\theta = \arccos(2x)$. Then $\cos\theta = 2x > 0$ and $\sin\theta = \sqrt{1-4x^2}$. Thus

$$\boxed{\frac{1}{5}\tan(\arccos(2x)) = \frac{\sqrt{1-4x^2}}{10x}}.$$

25. Simplify $\sec(\arcsin(2x))$.
Solution. Let $\theta = \arcsin(2x)$. Then $\cos\theta = \sqrt{1-4x^2}$. Consequently

$$\boxed{\sec(\arcsin(2x)) = \frac{1}{\sqrt{1-4x^2}}}, \qquad |x| < \frac{1}{2}.$$

4 Sign-Sensitive Compositions

These are the cases where the principal range can force an absolute value or a sign factor.

26. $\sin\left(\operatorname{arcsec}\frac{x}{3}\right)$

Solution. $\cos\theta = 3/x$ and $\theta \in [0,\pi]$ forces $\sin\theta \ge 0$: $\sin\theta = \sqrt{1-9/x^2} = \dfrac{\sqrt{x^2-9}}{|x|}$ for $|x| \ge 3$. The naive triangle gives $\sqrt{x^2-9}/x$, which is wrong for $x \le -3$.

27. $\tan\left(\operatorname{arcsec}\frac{x}{2}\right)$
Solution. $\cos\theta = 2/x$, $\sin\theta = \sqrt{x^2-4}/|x|$, so

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{x}{|x|}\cdot\frac{\sqrt{x^2-4}}{2} = \operatorname{sgn}(x)\,\frac{\sqrt{x^2-4}}{2}$$

for $|x| > 2$.

28. $\cot\left(\operatorname{arccsc}\frac{x}{5}\right)$
Solution. $\sin\theta = 5/x$ and $\theta \in [-\pi/2,\pi/2]$ forces $\cos\theta \ge 0$: $\cos\theta = \sqrt{x^2-25}/|x|$, so $\cot\theta = \operatorname{sgn}(x)\,\dfrac{\sqrt{x^2-25}}{5}$ for $|x| > 5$.

29. $\cos(\operatorname{arccsc} 3x)$

Solution. $\sin\theta = 1/(3x)$, $\cos\theta \ge 0$, so $\dfrac{\sqrt{9x^2-1}}{3|x|}$ for $|x| \ge 1/3$.

30. $\sec\left(\operatorname{arccsc}\dfrac{x}{2}\right)$

Solution. From the previous pattern $\cos\theta = \sqrt{x^2-4}/|x|$, so $\sec\theta = \dfrac{|x|}{\sqrt{x^2-4}}$ for $|x| > 2$.

31. $\csc(\operatorname{arcsec} 4x)$

Solution. $\cos\theta = 1/(4x)$, $\sin\theta = \sqrt{16x^2-1}/(4|x|) \geq 0$, so $\dfrac{4|x|}{\sqrt{16x^2-1}}$ for $|x| > 1/4$.

32. $\sin\left(\operatorname{arccot}\dfrac{x}{4}\right)$

Solution. $\cot\theta = x/4$ and $\theta \in (0,\pi)$ forces $\sin\theta > 0$: $\csc^2\theta = 1 + x^2/16$ gives $\csc\theta = \sqrt{16+x^2}/4$, so $\dfrac{4}{\sqrt{16+x^2}}$ for all x. The answer is positive even for $x < 0$.

33. $\cos\left(\operatorname{arccot}\dfrac{x}{4}\right)$

Solution. $\cos\theta = \cot\theta\sin\theta = \dfrac{x}{4}\cdot\dfrac{4}{\sqrt{16+x^2}} = \dfrac{x}{\sqrt{16+x^2}}$ for all x. Compare with the previous item: here the sign of x survives, there it does not.

5 Double-Angle Compositions

First recover the inner trigonometric quantities, then apply a double-angle identity.

34. $\sin\left(2\arcsin\dfrac{x}{3}\right)$

Solution. $2\sin\theta\cos\theta = 2\cdot\dfrac{x}{3}\cdot\dfrac{\sqrt{9-x^2}}{3} = \dfrac{2x\sqrt{9-x^2}}{9}$ for $|x| \leq 3$.

35. $\cos(2\arctan 2x)$

Solution. $\cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta} = \dfrac{1-4x^2}{1+4x^2}$ for all x.

36. $\sin\left(2\arctan\dfrac{x}{5}\right)$

Solution. $\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta} = \dfrac{2x/5}{1+x^2/25} = \dfrac{10x}{25+x^2}$ for all x.

37. $\tan\left(2\arctan\dfrac{x}{3}\right)$

Solution. $\dfrac{2\tan\theta}{1-\tan^2\theta} = \dfrac{2x/3}{1-x^2/9} = \dfrac{6x}{9-x^2}$ for $x \neq \pm 3$. The tangent is also undefined where $9-x^2=0$, the same points.

38. Simplify $2\sin(2\arctan x)$.

Solution. Put $\theta = \arctan x$. Then $\sin\theta = x/\sqrt{1+x^2}$ and $\cos\theta = 1/\sqrt{1+x^2}$. Hence

$$\boxed{2\sin(2\theta) = 4\sin\theta\cos\theta = \frac{4x}{1+x^2}}.$$

39. Simplify $\cos(2\arcsin(x/3))$.

Solution. Put $\theta = \arcsin(x/3)$. Using $\cos(2\theta) = 1 - 2\sin^2\theta$ gives

$$\boxed{\cos(2\arcsin(x/3)) = 1 - \frac{2x^2}{9}}, \qquad |x| \leq 3.$$

40. Simplify $\frac{1}{2}\tan(2\arctan(x/2))$.
Solution. Put $\theta = \arctan(x/2)$, so $\tan\theta = x/2$. Then

$$\frac{1}{2}\tan(2\theta) = \frac{1}{2}\frac{2(x/2)}{1-x^2/4} = \boxed{\frac{2x}{4-x^2}}, \qquad x \neq \pm 2.$$

At $x = \pm 2$ the original tangent is also undefined.

6 Final Mixed Review

Use whichever method is appropriate. The section title no longer tells which identity or sign issue is involved.

41. For $x \geq 3$, simplify $4\sin(\operatorname{arcsec}(x/3))$.
Solution. Let $\theta = \operatorname{arcsec}(x/3)$. Then $\cos\theta = 3/x$ and $\sin\theta \geq 0$. Therefore

$$\boxed{4\sin(\operatorname{arcsec}(x/3)) = \frac{4\sqrt{x^2-9}}{x}}.$$

42. For $x \neq 0$, simplify $\cot(\arctan(x/2))$.
Solution. If $\theta = \arctan(x/2)$, then $\tan\theta = x/2$. Thus

$$\boxed{\cot(\arctan(x/2)) = \frac{2}{x}}.$$

43. $\cos\left(2\operatorname{arcsec}\frac{x}{2}\right)$

Solution. $2\cos^2\theta - 1 = \dfrac{8}{x^2} - 1 = \dfrac{8-x^2}{x^2}$ for $|x| \geq 2$.

44. $\sin\left(2\operatorname{arcsec}\frac{x}{4}\right)$

Solution. $2\sin\theta\cos\theta = 2 \cdot \dfrac{\sqrt{x^2-16}}{|x|} \cdot \dfrac{4}{x} = \dfrac{8\,\operatorname{sgn}(x)\sqrt{x^2-16}}{x^2}$ for $|x| \geq 4$.

45. For $x \geq 2$, simplify $3\cos(2\operatorname{arcsec}(x/2))$.
Solution. Put $\theta = \operatorname{arcsec}(x/2)$, so $\cos\theta = 2/x$. Using $\cos(2\theta) = 2\cos^2\theta - 1$,

$$\boxed{3\cos(2\operatorname{arcsec}(x/2)) = 3\left(\frac{8}{x^2} - 1\right)}.$$

46. Simplify $\sin(2\arccos x)$.
Solution. Put $\theta = \arccos x$. Since $\theta \in [0, \pi]$, $\sin\theta = \sqrt{1-x^2}$, while $\cos\theta = x$. Thus

$$\boxed{\sin(2\arccos x) = 2x\sqrt{1-x^2}}, \qquad |x| \leq 1.$$

The factor x correctly supplies the sign when $\theta > \pi/2$.

Final check

For each answer, ask:

1. Is the answer entirely in terms of x?
2. Did I use the correct Pythagorean, reciprocal, or double-angle identity?
3. Did I choose the sign from the inverse function's principal range or the stated range of θ?
4. Does the final expression have the same domain as the original one?